

Homework 3

Electromagnetism - PHYS 715

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3.1 A spherical surface of radius R has charge uniformly distributed over its surface with a density $Q/4\pi R^2$, except for a spherical cap at the north pole, defined by the cone $\theta = \alpha$.

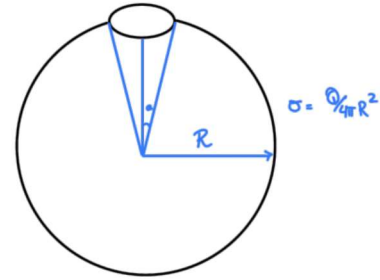
(a) Show that the potential inside the spherical surface can be expressed as

$$\Phi = \frac{Q}{8\pi\epsilon_0} \sum_{l=0}^{\infty} \frac{1}{2l+1} [P_{l+1}(\cos \alpha) - P_{l-1}(\cos \alpha)] \frac{r^l}{R^{l+1}} P_l(\cos \theta)$$

where, for $l=0$, $P_{l-1}(\cos \alpha) = -1$. What is the potential outside?

(b) Find the magnitude and the direction of the electric field at the origin.

(c) Discuss the limiting forms of the potential (part a) and electric field (part b) as the spherical cap becomes (1) very small, and (2) so large that the area with charge on it becomes a very small cap at the south pole.



a) There are many forms for Φ for which we can solve the problem.

We choose this expansion for $\frac{1}{|\vec{r} - \vec{r}'|} = \sum_{l=0}^{\infty} \frac{r_c^l}{r^{l+1}} P_l(\cos \gamma)$ where $\cos \gamma = \cos \theta \cos \theta' + \sin \theta \sin \theta' \cos(\phi - \phi')$

Alright, for $r < R$:

$$\Phi = \frac{1}{4\pi\epsilon_0} \int \frac{\sigma}{|\vec{r} - \vec{r}'|} dA'$$

Expanding $\frac{1}{|\vec{r} - \vec{r}'|}$ and replacing $\sigma = \frac{Q}{4\pi R^2}$, $dA' = R^2 \sin \theta' d\theta' d\phi'$

$$\begin{aligned} \Phi &= \frac{1}{4\pi\epsilon_0} \cdot \frac{Q}{4\pi R^2} \int_0^{2\pi} \int_0^\pi \sum_{l=0}^{\infty} \frac{r^l}{R^{l+1}} P_l(\cos \gamma) R^2 \sin \theta' d\theta' d\phi' \\ &= \frac{Q}{16\pi^2 \epsilon_0} \sum_{l=0}^{\infty} \frac{r^l}{R^{l+1}} \int_0^{2\pi} \int_0^\pi P_l(\cos \gamma) \sin \theta' d\theta' d\phi' \end{aligned}$$

We work with the ϕ' dependency first.

We use the identity (3.62) $P_l(\cos \gamma) = \frac{4\pi}{2l+1} \sum_{m=-l}^l Y_{lm}^*(\theta', \phi') Y_{lm}(\theta, \phi)$

& identity (3.53) $Y_{lm}(\theta, \phi) = \sqrt{\frac{2l+1}{4\pi} \frac{(l-m)!}{(l+m)!}} P_l^m(\cos \theta) e^{im\phi}$

& Fourier orthogonality $\int_0^{2\pi} e^{im\phi} d\phi = \begin{cases} 2\pi & m=0 \\ 0 & m \neq 0 \end{cases} \Rightarrow \int_0^{2\pi} e^{im\phi} d\phi = 2\pi \delta(m)$

$$\begin{aligned} \int_0^{2\pi} P_l(\cos \gamma) d\phi' &= \frac{4\pi}{2l+1} \sum_{m=-l}^l Y_{lm}(\theta, \phi) \int_0^{2\pi} Y_{lm}^*(\theta', \phi') d\phi' \\ &= \frac{4\pi}{2l+1} \sum_{m=-l}^l Y_{lm}(\theta, \phi) \int_0^{2\pi} \sqrt{\frac{2l+1}{4\pi} \frac{(l-m)!}{(l+m)!}} P_l^m(\cos \theta') e^{-im\phi'} d\phi' \\ &= \frac{4\pi}{2l+1} \sum_{m=-l}^l \sqrt{\frac{2l+1}{4\pi} \frac{(l-m)!}{(l+m)!}} P_l^m(\cos \theta) e^{im\phi} \sqrt{\frac{2l+1}{4\pi} \frac{(l-m)!}{(l+m)!}} P_l^m(\cos \theta') \cdot 2\pi \delta(m) \\ &= 2\pi P_l^0(\cos \theta) P_l^0(\cos \theta') \end{aligned}$$

all terms disappear except $m=0$.

We now work with θ' dependency.

We use the identity (3.28) $\frac{dP_{l+1}}{dx} - \frac{dP_{l-1}}{dx} - (2l+1)P_l = 0$

$$\begin{aligned} \int_a^\pi P_l(\cos \theta') \sin \theta' d\theta' &= - \int_{\cos a}^{-1} P_l(x) dx && \text{substitution of } x = \cos \theta' \\ &= \int_{-1}^{\cos a} P_l(x) dx \\ &= \frac{1}{2l+1} \int_{-1}^{\cos a} \left(\frac{dP_{l+1}}{dx} - \frac{dP_{l-1}}{dx} \right) dx \\ &= \frac{1}{2l+1} \left[P_{l+1}(\cos a) - P_{l-1}(\cos a) \right] - \left[P_{l+1}(-1) - P_{l-1}(-1) \right] \\ &= \frac{1}{2l+1} (P_{l+1}(\cos a) - P_{l-1}(\cos a)) && \begin{matrix} 0 \\ (-1)^{l+1} - (-1)^{l-1} \end{matrix} \end{aligned}$$

Putting Everything together

for $r < R$,

$$\begin{aligned} \Phi_{in} &= \frac{Q}{16\pi^2 \epsilon_0} \sum_{l=0}^{\infty} \frac{r^l}{R^{l+1}} \int_0^{2\pi} \int_a^\pi P_l(\cos \theta') \sin \theta' d\theta' d\theta \\ &= \frac{Q}{16\pi^2 \epsilon_0} \sum_{l=0}^{\infty} \frac{r^l}{R^{l+1}} \int_a^\pi 2\pi P_l(\cos \theta) P_l(\cos \theta') \sin \theta' d\theta' \\ &= \frac{Q}{16\pi^2 \epsilon_0} \sum_{l=0}^{\infty} \frac{r^l}{R^{l+1}} 2\pi P_l(\cos \theta) \frac{1}{2l+1} (P_{l+1}(\cos a) - P_{l-1}(\cos a)) \\ &= \frac{Q}{8\pi \epsilon_0} \sum_{l=0}^{\infty} \frac{1}{2l+1} \frac{r^l}{R^{l+1}} [P_{l+1}(\cos a) - P_{l-1}(\cos a)] P_l(\cos \theta) \end{aligned}$$

For $r > R$, we just flip r & R .

$$\Phi_{out} = \frac{Q}{8\pi \epsilon_0} \sum_{l=0}^{\infty} \frac{1}{2l+1} \frac{R^l}{r^{l+1}} [P_{l+1}(\cos a) - P_{l-1}(\cos a)] P_l(\cos \theta)$$

b) We need to find $E = -\nabla \Phi \big|_{r=0}$

Since we have azimuthal symmetry,

$$\begin{aligned} E \big|_{r=0} &= -\nabla \Phi \big|_{r=0} \\ &= -\left[\hat{r} \frac{\partial \Phi}{\partial r} + \hat{\theta} \frac{1}{r} \frac{\partial \Phi}{\partial \theta} \right] \bigg|_{r=0} \end{aligned}$$

Applying the derivative of the first term & dividing by r in the second term then evaluating for $r=0$. We find that $l=1$ is the only term that survives.

We also use the following Legendre Polynomials:

$$P_0(x) = 1$$

$$P_1(x) = x$$

$$P_2(x) = \frac{1}{2}(3x^2 - 1)$$

$$\begin{aligned}
E|_{r=0} &= -\left[\hat{r} \frac{Q}{8\pi\epsilon_0} \sum_{l=0}^{\infty} \frac{l}{2l+1} \frac{r^{l-1}}{R^{l+1}} \left[P_{l+1}(\cos a) - P_{l-1}(\cos a) \right] P_l(\cos \theta) + \hat{\theta} \frac{Q}{8\pi\epsilon_0} \sum_{l=0}^{\infty} \frac{1}{2l+1} \frac{r^{l-1}}{R^{l+1}} \left[P_{l+1}(\cos a) - P_{l-1}(\cos a) \right] P_l(\cos \theta) \right] \\
&= -\left(\hat{r} \frac{Q}{8\pi\epsilon_0} \frac{1}{3} \frac{1}{R^2} \left[P_2(\cos a) - P_0(\cos a) \right] P_1(\cos \theta) + \hat{\theta} \frac{Q}{8\pi\epsilon_0} \frac{1}{3} \frac{1}{R^2} \left[P_2(\cos a) - P_0(\cos a) \right] \frac{\partial}{\partial \theta} P_1(\cos \theta) \right) \\
&= \frac{Q}{24\pi\epsilon_0 R^2} \left[P_0(\cos a) - P_2(\cos a) \right] \left(\hat{r} P_1(\cos \theta) + \hat{\theta} \frac{\partial}{\partial \theta} P_1(\cos \theta) \right) \\
&= \frac{Q}{24\pi\epsilon_0 R^2} \left[1 - \frac{3\cos^2 a}{2} + \frac{1}{2} \right] \left(\hat{r} \cos \theta - \hat{\theta} \sin \theta \right) \\
&= \frac{Q}{24\pi\epsilon_0 R^2} \frac{3}{2} (1 - \cos^2 a) \hat{z} \\
&= \frac{Q}{16\pi\epsilon_0 R^2} \sin^2 a \hat{z}
\end{aligned}$$

c) We need to find the limiting behaviors - for which $a \rightarrow 0$ and $a \rightarrow \pi$ for Φ_n , Φ_{out} & $E|_{r=0}$.

We first start with $a \rightarrow 0$.

We realize we need to manipulate this term $P_{l+1}(\cos a) - P_{l-1}(\cos a)$.

$$\begin{aligned}
P_l(\cos a) &\approx P_l\left(1 - \frac{1}{2}a^2\right) && \text{Power Series of } \cos a = \sum_{n=0}^{\infty} \frac{(-1)^n}{(2n)!} a^{2n} = 1 - \frac{a^2}{2!} + \dots \\
&\approx P_l(1) + \left(-\frac{1}{2}a^2\right) P_l'(1) && \text{Power Series of } P_l \text{ around } x=1: P_l(x) = P_l(1) + (x-1)P_l'(1) + \dots \\
&= 1 - \frac{1}{2}a^2 P_l'(1) && P_l(1) = 1
\end{aligned}$$

$$\begin{aligned}
P_{l+1}(\cos a) - P_{l-1}(\cos a) &\approx -\frac{1}{2}a^2 \left(P_{l+1}'(1) - P_{l-1}'(1) \right) \\
&= -\frac{1}{2}a^2 \left((2l+1) P_l(1) \right) \\
&= -\frac{1}{2}a^2 (2l+1) && l \neq 0.
\end{aligned}$$

We must be very careful handling the $l=0$ case.

$$\begin{aligned}
P_1(\cos a) - P_{-1}(\cos a) &\approx 1 - \frac{1}{2}a^2 P_1'(1) - (-1) && \frac{d}{dx} P_1(x) = \frac{d}{dx} x = 1 \\
&= 2 - \frac{1}{2}a^2
\end{aligned}$$

Plugging in those values in Φ_n , Φ_{out} , $E|_{r=0}$.

$$\begin{aligned}
\Phi_n &= \frac{Q}{8\pi\epsilon_0} \sum_{l=0}^{\infty} \frac{1}{2l+1} \frac{r^l}{R^{l+1}} \left[P_{l+1}(\cos a) - P_{l-1}(\cos a) \right] P_l(\cos \theta) \\
&\approx \frac{Q}{8\pi\epsilon_0 R} \left(2 - \frac{1}{2}a^2 \right) P_0(\cos \theta) + \frac{Q}{8\pi\epsilon_0} \sum_{l=1}^{\infty} \frac{1}{2l+1} \frac{r^l}{R^{l+1}} \left(-\frac{1}{2}a^2 (2l+1) \right) P_l(\cos \theta) \\
&= \frac{Q}{4\pi\epsilon_0 R} - \frac{Qa^2}{16\pi\epsilon_0} \sum_{l=1}^{\infty} \frac{r^l}{R^{l+1}} P_l(\cos \theta) && \cos \theta = \cos \theta \cos \theta' + \sin \theta \sin \theta \cos(\theta - \theta') \Rightarrow \theta' = 0 \Rightarrow \hat{z} \\
&= \frac{Q}{4\pi\epsilon_0 R} - \frac{Qa^2}{16\pi\epsilon_0} \frac{1}{|\vec{r} - R\hat{z}|}
\end{aligned}$$

The Φ_{out} case can be found by switching r and R .

$$\begin{aligned}\Phi_{out} &= \frac{Q}{8\pi\epsilon_0} \sum_{l=0}^{\infty} \frac{1}{2l+1} \frac{R^l}{r^{l+1}} [P_{l+1}(\cos\alpha) - P_{l-1}(\cos\alpha)] P_l(\cos\theta) \\ &\approx \frac{Q}{8\pi\epsilon_0} \frac{1}{r} (2 - \frac{1}{2}a^2) P_0(\cos\theta) + \frac{Q}{8\pi\epsilon_0} \sum_{l=1}^{\infty} \frac{1}{2l+1} \frac{R^l}{r^{l+1}} \left(-\frac{1}{2}a^2(2l+1)\right) P_l(\cos\theta) \\ &= \frac{Q}{4\pi\epsilon_0 r} - \frac{Qa^2}{16\pi\epsilon_0} \sum_{l=0}^{\infty} \frac{R^l}{r^{l+1}} P_l(\cos\theta) \quad \cos\theta = \cos\theta \cos\theta' + \sin\theta \sin\theta' \cos(\theta - \theta') \Rightarrow \theta' = 0 \Rightarrow \hat{z} \\ &= \frac{Q}{4\pi\epsilon_0 r} - \frac{Qa^2}{16\pi\epsilon_0} \frac{1}{|\vec{r} - R\hat{z}|} \quad \frac{1}{|R\hat{z} - \vec{r}|} = \frac{1}{|\vec{r} - R\hat{z}|}\end{aligned}$$

We finish off with solution to $E|_{r=0}$ by using $\sin^2\alpha \approx a^2$

$$\begin{aligned}E|_{r=0} &= \frac{Q}{16\pi\epsilon_0 R^2} \sin^2\alpha \hat{z} \\ &\approx \frac{Qa^2}{16\pi\epsilon_0 R^2} \hat{z}\end{aligned}$$

Now, for the case of a small cap of charge $a \rightarrow \pi$.

We consider a_1 such that $a = \pi - a_1$, so that when $a \rightarrow \pi$, $a_1 \rightarrow 0$.

a_1 represent the angle of the small cap



$$\begin{aligned}P_l(\cos\alpha) &= P_l(\cos(\pi - a_1)) \\ &= P_l(-\cos a_1) \\ &= (-1)^l P_l(\cos a_1) \\ &\approx (-1)^l \left(1 - \frac{1}{2}a_1^2 P'_l(1)\right)\end{aligned}$$

$$\begin{aligned}P_{l+1}(\cos\alpha) - P_{l-1}(\cos\alpha) &\approx (-1)^{l+1} \left(1 - \frac{1}{2}a_1^2 P'_{l+1}(1)\right) - (-1)^{l-1} \left(1 - \frac{1}{2}a_1^2 P'_{l-1}(1)\right) \\ &= (-1)^{l-1} \left[-\frac{1}{2}a_1^2 (P'_{l+1}(1) - P'_{l-1}(1))\right] \\ &= (-1)^l \left[\frac{1}{2}a_1^2 (2l+1) P_l(1)\right] \\ &= \frac{1}{2}a_1^2 (-1)^l (2l+1) \quad (1)\end{aligned}$$

Checking the $l=0$ case

$$\begin{aligned}P_1(\cos\alpha) - P_{-1}(\cos\alpha) &= \cos\alpha + 1 \\ &= -\cos a_1 + 1 \\ &\approx -(1 - \frac{1}{2}a_1^2) + 1 \\ &\approx \frac{1}{2}a_1^2\end{aligned}$$

This follows the same rule as (1) so we don't need to split

We go through exactly the same process but with a factor of $(-1)^l$.

$$\Phi_{in} = \frac{Q}{8\pi\epsilon_0} \sum_{l=0}^{\infty} \frac{1}{2l+1} \frac{r^l}{R^{l+1}} \left[P_{l+1}(\cos\alpha) - P_{l-1}(\cos\alpha) \right] P_l(\cos\theta)$$

$$\approx \frac{Q}{8\pi\epsilon_0} \sum_{l=0}^{\infty} \frac{1}{2l+1} \frac{r^l}{R^{l+1}} \left(-(-1)^{\frac{l}{2}} \frac{1}{2} a^2 (2l+1) \right) P_l(\cos\theta)$$

$$= \frac{Qa^2}{16\pi\epsilon_0} \sum_{l=0}^{\infty} \frac{r^l}{R^{l+1}} P_l(-\cos\theta) \quad \cos\theta = \cos\theta \cos\theta' + \sin\theta \sin\theta' \cos(\theta - \theta') \Rightarrow \theta' = \pi, z = -\hat{z}$$

$$= \frac{Qa^2}{16\pi\epsilon_0} \frac{1}{|\vec{r} - (-R\hat{z})|}$$

The Φ_{out} case is the same but flipping r and R .

$$\Phi_{out} = \frac{Qa^2}{16\pi\epsilon_0} \frac{1}{|\vec{r} - (-R\hat{z})|} \quad \frac{1}{|\vec{r} - (-R\hat{z})|} = \frac{1}{|R\hat{z} - \vec{r}|}$$

We finally finish off with the solution to $E|_{r=0}$ using $\sin^2\alpha = \sin^2\pi - \alpha_1$
 $= \sin^2\alpha_1 \approx \alpha_1^2$

$$E|_{r=0} = \frac{Q}{16\pi\epsilon_0 R^2} \sin^2\alpha \hat{z}$$

$$\approx \frac{Q\alpha_1^2}{16\pi\epsilon_0 R^2} \hat{z}$$

3.4 A hollow sphere of inner radius a has the potential specified on its surface to be $\Phi = V(\theta, \phi)$. Prove the equivalence of the two forms of solution for the potential inside the sphere:

$$(a) \Phi(x) = \frac{a(a^2 - r^2)}{4\pi} \int \frac{V(\theta', \phi')}{(r^2 + a^2 - 2ar \cos \gamma)^{3/2}} d\Omega'$$

where $\cos \gamma = \cos \theta \cos \theta' + \sin \theta \sin \theta' \cos(\phi - \phi')$.

$$(b) \Phi(x) = \sum_{l=0}^{\infty} \sum_{m=-1}^l A_{lm} \left(\frac{r}{a}\right)^l Y_{lm}(\theta, \phi)$$

where $A_{lm} = \int d\Omega' Y_{lm}^*(\theta', \phi') V(\theta', \phi')$.

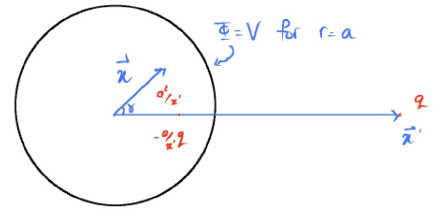
Part a is the solution provided by method of images & Part b is the solution by the spherical harmonics.

Let's prove both & show they are equivalent:-

a) This is equivalent to (2.19) but we find the solution inside the sphere.

Let q be a unit point source at x' .

Then, the solution to Green's function is:



$$G(\vec{x}, \vec{x}') = \frac{1}{|\vec{x} - \vec{x}'|} - \frac{q/x'}{|\vec{x} - \frac{a^2}{x'^2} \vec{x}'|}$$

Writing in spherical coordinates

$$G(\vec{x}, \vec{x}') = \frac{1}{\sqrt{x^2 + x'^2 - 2xx' \cos \gamma}} - \frac{q/x'}{\sqrt{x^2 + \frac{a^4}{x'^2} - 2x \frac{a^2}{x'} \cos \gamma}}$$

$$= \frac{1}{\sqrt{x^2 + x'^2 - 2xx' \cos \gamma}} - \frac{1}{\sqrt{\frac{x'^2 x^2}{a^2} + a^2 - 2xx' \cos \gamma}}$$

We see that the Dirichlet condition is satisfied $\Rightarrow G=0$ if $\vec{x}, \vec{x}'$ on S

We also require $\frac{\partial G}{\partial n'} \Big|_{x'=a}$ to find the solution to Φ (1.44) i.e. $x=a$ or $x'=a$

$$\frac{\partial G}{\partial n'} \Big|_{x'=a} = \frac{-\frac{1}{2}(2x' - 2x \cos \gamma)}{\sqrt{x^2 + x'^2 - 2xx' \cos \gamma}^3} - \frac{-\frac{1}{2}\left(\frac{2x'x^2}{a^2} - 2x \cos \gamma\right)}{\sqrt{\frac{x'^2 x^2}{a^2} + a^2 - 2xx' \cos \gamma}^3} \Big|_{x'=a}$$

$$= \frac{-a + x \cos \gamma}{\sqrt{x^2 + a^2 - 2xa \cos \gamma}^3} - \frac{-\frac{x^2}{a} + x \cos \gamma}{\sqrt{x^2 + a^2 - 2xa \cos \gamma}^3}$$

$$= \frac{x^2 - a^2}{a \sqrt{x^2 + a^2 - 2xa \cos \gamma}^3}$$

We now go back to the solution to Φ by the Dirichlet Boundary Conditions:-

$$\Phi(x) = \underbrace{\frac{1}{4\pi\epsilon_0} \int_V \rho(\vec{x}') G_p(\vec{x}, \vec{x}') d^3x'}_{\text{no charge in } V} - \frac{1}{4\pi} \oint_S \Phi(x') \frac{\partial G_0}{\partial n'} da'$$

$$= \frac{1}{4\pi} \int \frac{a^2 - x^2}{a \sqrt{x^2 + a^2 - 2xa \cos \gamma}^3} \cdot \Phi(a, \theta', \phi') \cdot a^2 d\Omega'$$

$$= \frac{a(a^2 - x^2)}{4\pi} \int \frac{V(\theta', \phi')}{\sqrt{x^2 + a^2 - 2xa \cos \gamma}^3} d\Omega'$$

b) Let's prove the solution to Φ using spherical harmonics.

$$\Phi = \sum_{l=0}^{\infty} \sum_{m=-l}^l [A_{lm} r^l + B_{lm} r^{-l-1}] Y_{lm}(\theta, \phi)$$

Since we are solving for inside the sphere, then Φ must be finite at $r=0 \Rightarrow B_{lm} = 0$ since as $r \rightarrow 0$, $r^{-l-1} \rightarrow \infty$

$$\Phi = \sum_{l=0}^{\infty} \sum_{m=-l}^l A_{lm} r^l Y_{lm}(\theta, \phi)$$

We now apply the boundary condition $\Phi(a, \theta, \phi) = V(\theta, \phi)$ to find coefficients A_{lm} .

$$V(\theta, \phi) = \sum_{l=0}^{\infty} \sum_{m=-l}^l A_{lm} a^l Y_{lm}(\theta, \phi)$$

Applying the inverse Fourier transform:

$$\iint Y_{l'm'}^*(\theta', \phi') V(\theta', \phi') d\Omega' = \sum_{l=0}^{\infty} \sum_{m=-l}^l A_{lm} a^l \underbrace{\iint Y_{l'm'}^* Y_{lm} d\Omega'}_{\substack{\oint_{\Omega'} \oint_{\Omega''} \text{ since } Y_{l'm'}^* \perp Y_{lm}}}$$

$$a^{-l} \iint Y_{l'm'}^*(\theta', \phi') V(\theta', \phi') d\Omega' = A_{lm}$$

We finish off the solution to Φ by plugging in A_{lm} .

$$\Phi = \sum_{l=0}^{\infty} \sum_{m=-l}^l A_{lm} \left(\frac{r}{a}\right)^l Y_{lm}(\theta, \phi) \quad \text{where } A_{lm} = \int Y_{lm}^*(\theta', \phi') V(\theta', \phi') d\Omega'$$

We can also prove equivalence of both forms by analyzing the Green's function in spherical harmonics.

$$\frac{1}{|\vec{r} - \vec{r}'|} = 4\pi \sum_{l=0}^{\infty} \sum_{m=-l}^l \frac{1}{2l+1} \frac{r^l}{a^{l+1}} Y_{lm}^*(\theta', \phi') Y_{lm}(\theta, \phi) \quad (3.70) \quad \text{since } r < a.$$

Examine the term $\frac{1}{2l+1} \frac{r^l}{a^{l+1}}$ from Green's and $\frac{r^l}{a^l}$ in the potential Φ from last part.

If we can get rid of $2l+1$ we are set!

$$\begin{aligned} \frac{1}{2l+1} \frac{r^l}{a^{l+1}} &\xrightarrow{\partial_r} \frac{l}{2l+1} \frac{r^{l-1}}{a^{l+1}} \xrightarrow{r} \frac{l}{2l+1} \frac{r^l}{a^{l+1}} \\ &\xrightarrow{\partial_a} \frac{-l+1}{2l+1} \frac{r^l}{a^{l+2}} \xrightarrow{a} \frac{l+1}{2l+1} \frac{r^l}{a^{l+1}} \end{aligned} \quad \left. \vphantom{\frac{1}{2l+1} \frac{r^l}{a^{l+1}}} \right\} \text{We note that adding these terms and multiplying by } a \text{ gives us our desired result } \frac{r^l}{a^l}!$$

Therefore, let us conduct the operator $a(r\partial_r - a\partial_a)$ on Green's function and see what we get.

$$\begin{aligned} \frac{1}{|\vec{r}_a - \vec{r}|} &= \frac{1}{\sqrt{a^2 + r^2 - 2ar\cos\gamma}} \xrightarrow{r\partial_r} \frac{-\frac{1}{2}(2r^2 - 2ar\cos\gamma)}{\sqrt{a^2 + r^2 - 2ar\cos\gamma}^3} \\ &\xrightarrow{-a\partial_a} \frac{-\frac{1}{2}(-2a^2 + 2ar\cos\gamma)}{\sqrt{a^2 + r^2 - 2ar\cos\gamma}^3} \\ &\xrightarrow{a(r\partial_r - a\partial_a)} \frac{a(a^2 - r^2)}{\sqrt{a^2 + r^2 - 2ar\cos\gamma}^3} \quad (1) \end{aligned}$$

Recall the method of images solution to Φ is

$$\Phi(x) = \frac{a(a^2 - x^2)}{4\pi} \int \frac{V(\theta', \phi')}{\sqrt{x^2 + a^2 - 2xa\cos\gamma}^3} d\Omega'$$

$$= \frac{1}{4\pi} \int a(r\partial_r - a\partial_a) \left[\frac{1}{|\vec{r}_a - \vec{r}|} \right] V(\theta', \phi') d\Omega'$$

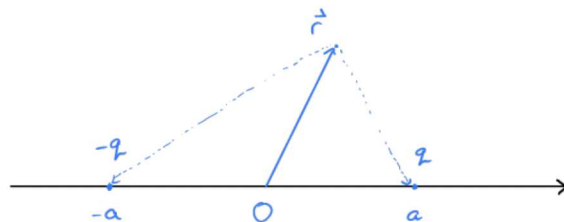
$$= \frac{1}{4\pi} \int 4\pi \sum_{l=0}^{\infty} \sum_{m=-l}^l \frac{r^l}{a^{l+1}} Y_{lm}^*(\theta', \phi') Y_{lm}(\theta, \phi) V(\theta', \phi') d\Omega'$$

$$= \sum_{l=0}^{\infty} \sum_{m=-l}^l A_{lm} \left(\frac{r}{a} \right)^l Y_{lm}(\theta, \phi) \quad \text{where } A_{lm} = \int Y_{lm}^*(\theta', \phi') V(\theta', \phi') d\Omega'$$

□

3.5 Two point charges q and $-q$ are located on the z axis at $z = +a$ and $z = -a$, respectively.

- (a) Find the electrostatic potential as an expansion in spherical harmonics and powers of r for both $r > a$ and $r < a$.
 (b) Keeping the product $qa \equiv p/2$ constant, take the limit of $a \rightarrow 0$ and find the potential for $r \neq 0$. This is by definition a dipole along the z axis and its potential.
 (c) Suppose now that the dipole of part b is surrounded by a grounded spherical shell of radius b concentric with the origin. By linear superposition find the potential everywhere inside the shell.



The potential Φ at point $\vec{r}$ from charges at $z = -a$ and $z = a$ is

$$\Phi(\vec{r}) = \frac{q/4\pi\epsilon_0}{|\vec{r} - \hat{k}a|} - \frac{q/4\pi\epsilon_0}{|\vec{r} + \hat{k}a|}$$

We proceed by expanding the denominator in spherical harmonics

$$\frac{1}{|\vec{r} - \vec{r}_0|} = 4\pi \sum_{l=0}^{\infty} \sum_{m=-l}^l \frac{1}{2l+1} \frac{r_{<}^l}{r_{>}^{l+1}} Y_{lm}^*(\theta', \phi') Y_{lm}(\theta, \phi) \quad (3.70)$$

The coordinates (z', θ', ϕ') for $a\hat{k}$ are $(a, 0, 0)$ & $-a\hat{k}$ are $(-a, \pi, 0)$

Therefore,

$$\Phi(\vec{r}) = \frac{q}{4\pi\epsilon_0} \left(4\pi \sum_{l=0}^{\infty} \sum_{m=-l}^l \frac{1}{2l+1} \frac{r_{<}^l}{r_{>}^{l+1}} Y_{lm}(\theta, \phi) [Y_{lm}^*(0, 0) - Y_{lm}^*(\pi, 0)] \right) \quad \text{for } r_{<} \text{ is for which ever is less between } r \text{ & } a$$

We can actually do simplify the expression further since we notice that m must be 0

1) $Y_{lm}(0, 0) = Y_{lm}(\pi, 0) = 0$ for $m \neq 0$ since they contain $\sin \theta$ terms

2) The problem has azimuthal symmetry and the only term with no ϕ dependence is Y_{l0}

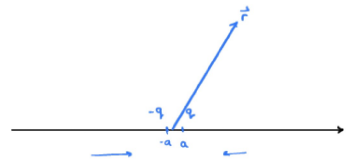
$$\begin{aligned} \Phi(\vec{r}) &= \frac{q}{4\pi\epsilon_0} \left(4\pi \sum_{l=0}^{\infty} \frac{1}{2l+1} \frac{r_{<}^l}{r_{>}^{l+1}} Y_{l0}(\theta, \phi) [Y_{l0}^*(0, 0) - Y_{l0}^*(\pi, 0)] \right) \\ &= \frac{q}{4\pi\epsilon_0} \left(4\pi \sum_{l=0}^{\infty} \frac{1}{2l+1} \frac{r_{<}^l}{r_{>}^{l+1}} \sqrt{\frac{2l+1}{4\pi}} P_l(\cos \theta) \cdot \sqrt{\frac{2l+1}{4\pi}} (P_l(\cos 0) - P_l(\cos \pi)) \right) \quad Y_{l0} = \sqrt{\frac{2l+1}{4\pi}} P_l(\cos \theta) \quad (3.57) \\ &= \frac{q}{4\pi\epsilon_0} \left(\sum_{l=0}^{\infty} \frac{r_{<}^l}{r_{>}^{l+1}} P_l(\cos \theta) [P_l(1) - P_l(-1)] \right) \\ &= \frac{q}{4\pi\epsilon_0} \sum_{l=0, \text{ odd}}^{\infty} \frac{r_{<}^l}{r_{>}^{l+1}} P_l(\cos \theta) \cdot 2 \end{aligned}$$

if $r < a$ then $r_{<} = r$, $r_{>} = a$

and $r > a$ then $r_{<} = a$, $r_{>} = r$.

b) If $a \rightarrow 0$, then we are investigating the region for $r > a$.

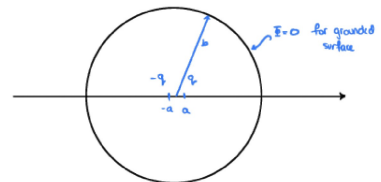
$$\begin{aligned}\Phi(\vec{r}) &= \frac{2q}{4\pi\epsilon_0} \sum_{l=0, \text{ odd}}^{\infty} \frac{a^l}{r^{l+1}} P_l(\cos\theta) \\ &= \frac{2qa}{4\pi\epsilon_0} \sum_{l=0, \text{ odd}}^{\infty} \frac{a^{l-1}}{r^{l+1}} P_l(\cos\theta) \\ &= \frac{P}{4\pi\epsilon_0} \left[\frac{1}{r^2} P_1(\cos\theta) + \frac{a^2}{r^4} P_3(\cos\theta) + \dots \right]\end{aligned}$$



For $a \rightarrow 0$, all higher terms of a go to zero and we are left with:

$$\Phi(\vec{r}) = \frac{P}{4\pi\epsilon_0 r^2} \cos\theta \rightarrow \text{this is the potential for a perfect dipole along the z-axis (1.25)}$$

c) We are investigating the solution for potential Φ with a dipole in the centre of a grounded sphere.



We note:

1) $\Phi = 0$ for $r = b$ since it's grounded.

2) $\Phi = \Phi_{\text{dipole}} + \Phi_{\text{induced from sphere}}$

3) $\Phi_{\text{induced from sphere}} = \sum_{l=0}^{\infty} A_l r^l P_l(\cos\theta)$ since $m=0$ & $P_l=0$ (azimuthal symmetry and solution in sphere)

$$\Phi = \Phi_{\text{dipole}} + \Phi_{\text{induced from sphere}}$$

Applying the boundary conditions:

$$= \frac{P}{4\pi\epsilon_0 r^2} \cos\theta + \sum_{l=0}^{\infty} A_l r^l P_l(\cos\theta)$$

$$0 = \frac{P}{4\pi\epsilon_0 b^2} \cos\theta + \sum_{l=0}^{\infty} A_l b^l P_l(\cos\theta)$$

We examine that the only term that has $\cos\theta$ is $P_1(\cos\theta)$ i.e. $l=1$.

Therefore for $l=1$,

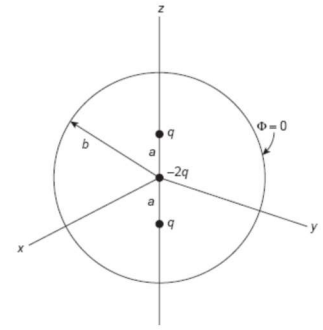
$$0 = \frac{P}{4\pi\epsilon_0 b^2} \cos\theta + A_1 b \cos\theta$$

$$\Rightarrow A_1 = -\frac{P}{4\pi\epsilon_0 b^3} \quad \therefore$$

$$\Phi(\vec{r}) = \frac{P}{4\pi\epsilon_0 r^2} \cos\theta - \frac{P}{4\pi\epsilon_0 b^3} r \cos\theta$$

$$= \frac{P}{4\pi\epsilon_0} \cos\theta \left(\frac{1}{r^2} - \frac{r}{b^3} \right) \quad \square$$

3.6 Three point charges ($q, -2q, q$) are located in a straight line with separation a and with the middle charge ($-2q$) at the origin of a grounded conducting spherical shell of radius b , as indicated in the sketch.



- (a) Write down the potential of the three charges in the absence of the grounded sphere. Find the limiting form of the potential as $a \rightarrow 0$, but the product $qa^2 = Q$ remains finite. Write this latter answer in spherical coordinates.
- (b) The presence of the grounded sphere of radius b alters the potential for $r < b$. The added potential can be viewed as caused by the surface-charge density induced on the inner surface at $r = b$ or by image charges located at $r > b$. Use linear superposition to satisfy the boundary conditions and find the potential everywhere inside the sphere for $r < a$ and $r > a$. Show that in the limit $a \rightarrow 0$,

$$\Phi(r, \theta, \phi) \rightarrow \frac{Q}{2\pi\epsilon_0 r^3} \left(1 - \frac{r^5}{b^5}\right) P_2(\cos \theta)$$

We will use the same techniques as 3.5 to solve this problem noting that we are now working with a quadrupole rather than a dipole.

- a) Using linear superposition, we find that the potential Φ from all charges is

$$\Phi(\vec{r}) = \frac{q}{4\pi\epsilon_0} \frac{1}{|\vec{r} - a\hat{k}|} + \frac{q}{4\pi\epsilon_0} \frac{1}{|\vec{r} + a\hat{k}|} - \frac{2q}{4\pi\epsilon_0} \frac{1}{|\vec{r}|}$$

We can then expand the first two terms in terms of the Legendre polynomials as in 3.5.

$$\frac{1}{|\vec{r} - \vec{r}_0|} = \sum_{l=0}^{\infty} \frac{r_0^l}{r^{l+1}} P_l(\cos \theta) P_l(\cos \theta')$$

For $r > a$,

$$\begin{aligned} \Phi(\vec{r}) &= \frac{q}{4\pi\epsilon_0} \sum_{l=0}^{\infty} \frac{a^l}{r^{l+1}} P_l(\cos \theta) \cdot 1 + \frac{q}{4\pi\epsilon_0} \sum_{l=0}^{\infty} \frac{a^l}{r^{l+1}} P_l(\cos \theta) \cdot (-1)^l - \frac{2q}{4\pi\epsilon_0} \frac{1}{r} \\ &= \frac{q}{4\pi\epsilon_0} \left[-\frac{2}{r} + \sum_{l=0}^{\infty} \frac{a^l}{r^{l+1}} P_l(\cos \theta) [1 + (-1)^l] \right] \\ &= \frac{2q}{4\pi\epsilon_0} \left[-\frac{1}{r} + \frac{1}{r} + \sum_{l=2,4,\dots}^{\infty} \frac{a^l}{r^{l+1}} P_l(\cos \theta) \right] \\ &= \frac{2q}{4\pi\epsilon_0} \sum_{\substack{l=2,4,\dots \\ \text{even}}}^{\infty} \frac{a^l}{r^{l+1}} P_l(\cos \theta) \end{aligned}$$

For $r < a$,

$$\begin{aligned}\Phi(\vec{r}) &= \frac{q}{4\pi\epsilon_0} \sum_{l=0}^{\infty} \frac{r^l}{a^{l+1}} P_l(\cos\theta) \cdot 1 + \frac{q}{4\pi\epsilon_0} \sum_{l=0}^{\infty} \frac{r^l}{a^{l+1}} P_l(\cos\theta) \cdot (-1)^l - \frac{2q}{4\pi\epsilon_0} \frac{1}{r} \\ &= \frac{2q}{4\pi\epsilon_0} \left[\sum_{\substack{l=0,2,\dots \\ \text{even}}}^{\infty} \frac{r^l}{a^{l+1}} P_l(\cos\theta) - \frac{1}{r} \right]\end{aligned}$$

Alright, let's look at the limiting case for $a \rightarrow 0$ such that $Q = qa^2$ is finite

We are concerned therefore about the $r > a$ case &

$$\begin{aligned}\Phi(\vec{r}) &= \frac{2q}{4\pi\epsilon_0} \sum_{\substack{l=2,4,\dots \\ \text{even}}}^{\infty} \frac{a^l}{r^{l+1}} P_l(\cos\theta) \\ &= \frac{2Q}{4\pi\epsilon_0} \sum_{\substack{l=2,4,\dots \\ \text{even}}}^{\infty} \frac{a^{l-2}}{r^{l+1}} P_l(\cos\theta)\end{aligned}$$

For $a \rightarrow 0$, only the $l=2$ term survives &

$$\begin{aligned}\Phi(\vec{r}) &= \frac{2Q}{4\pi\epsilon_0} \frac{1}{r^3} P_2(\cos\theta) \\ &= \frac{Q}{4\pi\epsilon_0} \frac{1}{r^3} (3\cos^2\theta - 1)\end{aligned}$$

Aha! Notice that both the $l=0$ (monopole) & $l=1$ (dipole) terms vanish and we are only left with $l=2$ representing the quadrupole moment!

b) The solution is pretty similar to part c in the previous question.

We are investigating the solution for potential Φ with

a q_{quad} in the centre of a grounded sphere.

We note &

1) $\Phi = 0$ for $r = b$ since it's grounded.

2) $\Phi = \Phi_{\text{quadrupole}} + \Phi_{\text{induced from sphere}}$

3) $\Phi_{\text{induced from sphere}} = \sum_{l=0}^{\infty} q_l r^l P_l(\cos\theta)$ since $m=0$ & $P_l=0$ (azimuthal symmetry and solution in sphere)

for $r > a$,

$$\Phi = \Phi_{\text{quadrupole}} + \Phi_{\text{induced from sphere}}$$

$$= \frac{2q}{4\pi\epsilon_0} \sum_{\substack{l=2,4,\dots \\ \text{even}}}^{\infty} \frac{a^l}{r^{l+1}} P_l(\cos\theta) + \sum_{l=0}^{\infty} A_l r^l P_l(\cos\theta)$$

Applying the boundary conditions $\Rightarrow \Phi(r=b) = 0$

$$0 = \frac{2q}{4\pi\epsilon_0} \sum_{\substack{l=2,4,\dots \\ \text{even}}}^{\infty} \frac{a^l}{b^{l+1}} P_l(\cos\theta) + \sum_{l=0}^{\infty} A_l b^l P_l(\cos\theta)$$

$$A_l = -\frac{2q}{4\pi\epsilon_0} \cdot \frac{a^l}{b^{2l+1}} \quad \text{for } l=2,4,\dots \text{ even and } 0 \text{ otherwise}$$

$$\Phi(\vec{r}) = \sum_{\substack{l=2,\dots \\ \text{even}}}^{\infty} \frac{2q}{4\pi\epsilon_0} a^l \left(\frac{1}{r^{l+1}} - \frac{r^l}{b^{2l+1}} \right) P_l(\cos\theta) \quad \text{for } r > a.$$

Similarly, for $r < a$

$$\Phi = \Phi_{\text{quadrupole}} + \Phi_{\text{induced from sphere}}$$

$$= \frac{2q}{4\pi\epsilon_0} \left[\sum_{\substack{l=0,2,\dots \\ \text{even}}}^{\infty} \frac{r^l}{a^{l+1}} P_l(\cos\theta) - \frac{1}{r} \right] + \sum_{l=0}^{\infty} A_l r^l P_l(\cos\theta)$$

Applying the boundary conditions $\Rightarrow \Phi(r=b) = 0$

$$0 = \frac{2q}{4\pi\epsilon_0} \left[\sum_{\substack{l=0,2,\dots \\ \text{even}}}^{\infty} \frac{b^l}{a^{l+1}} P_l(\cos\theta) - \frac{1}{b} \right] + \sum_{l=0}^{\infty} A_l b^l P_l(\cos\theta)$$

$$A_l = -\frac{2q}{4\pi\epsilon_0} \cdot \frac{1}{a^{l+1}} \quad \text{for } l=2,4,\dots \text{ even}$$

$$A_0 = -\frac{2q}{4\pi\epsilon_0} \left(\frac{1}{a} - \frac{1}{b} \right) \quad \text{and } A_l = 0 \text{ otherwise}$$

$$\Phi(\vec{r}) = \frac{2q}{4\pi\epsilon_0} \left[\sum_{\substack{l=0,2,\dots \\ \text{even}}}^{\infty} \frac{r^l}{a^{l+1}} P_l(\cos\theta) - \frac{1}{r} \right] - \sum_{\substack{l=2,4 \\ \text{even}}}^{\infty} \frac{2q}{4\pi\epsilon_0} \frac{r^l}{a^{l+1}} P_l(\cos\theta) - \frac{2q}{4\pi\epsilon_0} \left(\frac{1}{a} - \frac{1}{b} \right) \quad (r < a)$$

Looking at the limiting case again for $a \rightarrow 0$, we are only concerned with region $r > a$:

$$\begin{aligned} \Phi(\vec{r}) &= \frac{2Q}{4\pi\epsilon_0} \frac{1}{r^3} P_2(\cos\theta) - \lim_{a \rightarrow 0} \sum_{\substack{l=2,\dots \\ \text{even}}}^{\infty} \frac{2Q}{4\pi\epsilon_0} a^{l-2} \frac{r^l}{b^{2l+1}} P_l(\cos\theta) \\ &= \frac{2Q}{4\pi\epsilon_0} P_2(\cos\theta) \left(\frac{1}{r^3} - \frac{r^2}{b^5} \right) \end{aligned}$$

$\hookrightarrow \frac{\Phi_{\text{induced}}}{a^2}$ ($r > a$) from $Q = qa^2$

$\hookrightarrow$ only $l=2$ survives

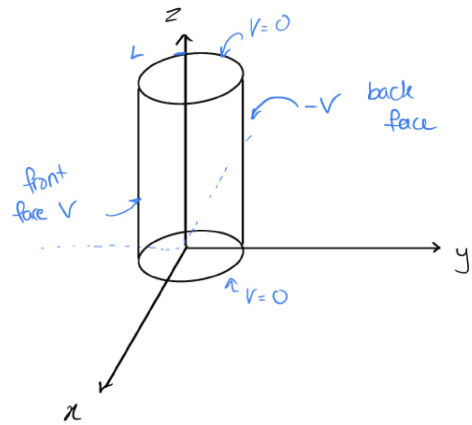
□

3.8 For the cylinder in [Worked Problem 3.2](#) the cylindrical surface is made of two equal half-cylinders, one at potential V and the other at potential $-V$, so that

$$V(\phi, z) = \begin{cases} V & \text{for } -\pi/2 < \phi < \pi/2 \\ -V & \text{for } \pi/2 < \phi < 3\pi/2 \end{cases}$$

(a) Find the potential inside the cylinder.

(b) Assuming $L \gg b$, consider the potential at $z = L/2$ as a function of ρ and ϕ .



a) We already have an explicit form for Φ in Worked Problem 3.2.

All we need is to find the appropriate coefficients

$$\Phi(\rho, \phi, z) = \sum_m \sum_n I_m\left(\frac{n\pi}{L}\rho\right) \sin\left(\frac{n\pi}{L}z\right) (B_{mn} \sin m\phi + A_{mn} \cos m\phi)$$

Applying the boundary condition at $\rho = b$.

$$V(\phi, z) = \sum_m \sum_n I_m\left(\frac{n\pi}{L}b\right) \sin\left(\frac{n\pi}{L}z\right) (B_{mn} \sin m\phi + A_{mn} \cos m\phi)$$

$$\begin{aligned} B_{mn} &= \frac{2}{\pi L I_m\left(\frac{n\pi}{L}b\right)} \left[\int_{-\pi/2}^{\pi/2} \sin m\phi \, d\phi \int_0^L V \sin\left(\frac{n\pi}{L}z\right) dz - \int_{\pi/2}^{3\pi/2} \sin m\phi \, d\phi \int_0^L V \sin\left(\frac{n\pi}{L}z\right) dz \right] \\ &= \frac{2}{\pi L I_m\left(\frac{n\pi}{L}b\right)} \left[-\frac{1}{m} \cos m\phi \Big|_{-\pi/2}^{\pi/2} \cdot -V \frac{L}{n\pi} \cos\left(\frac{n\pi}{L}z\right) \Big|_0^L - -\frac{1}{m} \cos m\phi \Big|_{\pi/2}^{3\pi/2} \cdot -V \frac{L}{n\pi} \cos\left(\frac{n\pi}{L}z\right) \Big|_0^L \right] \end{aligned}$$

We investigate further these cosine values.

$$\phi = -\pi/2, \quad \cos(-\pi/2 m) = \cos(\pi/2 m)$$

$$\begin{aligned} \phi = \pi/2, \quad \cos(\pi/2 m) &= 0 \quad \text{odd } m \\ &= (-1)^{m/2} \quad \text{even } m \end{aligned}$$

$$\begin{aligned} \phi = 3\pi/2, \quad \cos(3\pi/2 m) &= \cos \pi m \cos \pi/2 m - \sin \pi m \sin \pi/2 m = 0 \quad \text{odd } m \\ &= (-1)^m (-1)^{m/2} \quad \text{even } m \end{aligned}$$

Returning back:

$$B_{mn} = \frac{2}{\pi L I_m\left(\frac{n\pi}{L}b\right)} \left(-\frac{1}{m} \cos m\phi \Big|_{-\pi/2}^{\pi/2} \cdot -V \frac{L}{n\pi} \cos\left(\frac{n\pi}{L}z\right) \Big|_0^L - \frac{1}{m} \left[(-1)^{3m/2} - (-1)^{m/2} \right] V \frac{L}{n\pi} ((-1)^n - 1) \right)$$

$$= 0$$

We now solve for A_{mn} :-

$$A_{mn} = \frac{2}{\pi L I_m(\frac{n\pi b}{L})} \left[\int_{-\pi/2}^{\pi/2} \cos m\phi \, d\phi \int_0^L V \sin(\frac{n\pi}{L} z) \, dz - \int_{\pi/2}^{3\pi/2} \cos m\phi \, d\phi \int_0^L V \sin(\frac{n\pi}{L} z) \, dz \right]$$

$$= \frac{2}{\pi L I_m(\frac{n\pi b}{L})} \left[\frac{1}{m} \sin m\phi \Big|_{-\pi/2}^{\pi/2} \cdot -V \frac{L}{n\pi} \cos(\frac{n\pi}{L} z) \Big|_0^L - \frac{1}{m} \sin m\phi \Big|_{\pi/2}^{3\pi/2} \cdot -V \frac{L}{n\pi} \cos(\frac{n\pi}{L} z) \Big|_0^L \right]$$

We investigate further these sin values.

$$\phi = -\pi/2, \quad \sin(-\pi/2 m) = -\sin(\pi/2 m)$$

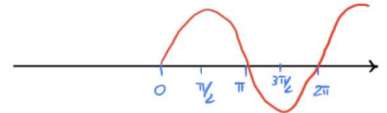
$$\phi = \pi/2, \quad \sin(\pi/2 m) = 0 \text{ for even } m$$

$$= (-1)^{m-1/2} \text{ for odd } m$$

$$\phi = 3\pi/2, \quad \sin(3\pi/2 m) = \sin \pi m \cos \pi/2 m + \cos \pi m \sin \pi/2 m$$

$$= 0 \text{ for even } m$$

$$= (-1)^m (-1)^{m-1/2} \text{ for odd } m$$



Going back to A_{mn}

odd m

$$A_{mn} = \frac{2}{\pi L I_m(\frac{n\pi b}{L})} \left[\frac{1}{m} \cdot 2(-1)^{m-1/2} \cdot \frac{-VL}{n\pi} [(-1)^n - 1] - \frac{1}{m} (-1)^{m-1/2} [(-1)^n - 1] \cdot \frac{-VL}{n\pi} [(-1)^n - 1] \right]$$

$$= \frac{2}{\pi L I_m(\frac{n\pi b}{L})} \cdot \frac{-VL}{mn\pi} (-1)^{m-1/2} \underbrace{[(-1)^n - 1]}_{-2 \text{ } n=\text{odd}} \underbrace{[2 + 1 - (-1)^n]}_{4 \text{ } m=\text{odd}}$$

$$= \frac{16V}{mn\pi^2 I_m(\frac{n\pi b}{L})} (-1)^{m-1/2} \text{ for } \begin{matrix} n = \text{odd} \\ m = \text{odd} \end{matrix} \quad \text{else } 0 \text{ otherwise}$$

Plugging in A_{mn} in our equation for Φ

$$\Phi(\rho, \phi, z) = \sum_{\substack{m \\ \text{odd}}} \sum_{\substack{n \\ \text{odd}}} I_m(\frac{n\pi}{L} \rho) \sin(\frac{n\pi}{L} z) (B_{mn} \sin m\phi + A_{mn} \cos m\phi)$$

$$= \frac{16V}{\pi^2} \sum_{\substack{m \\ \text{odd}}} \sum_{\substack{n \\ \text{odd}}} \frac{I_m(\frac{n\pi}{L} \rho)}{I_m(\frac{n\pi}{L} b)} \sin(\frac{n\pi}{L} z) \frac{(-1)^{m-1/2}}{mn} \cos m\phi$$

b) Assuming that $L \gg b$ i.e. $\frac{b}{L} \ll 1$ & considering the potential at $z = \frac{L}{2}$

$$\Phi(\rho, \phi) = \frac{16V}{\pi^2} \sum_{m \text{ odd}} \sum_{n \text{ odd}} \frac{I_m\left(\frac{n\pi}{L}\rho\right)}{I_m\left(\frac{n\pi}{L}b\right)} \sin\left(\frac{n\pi}{2}\right) \frac{(-1)^{m-1/2}}{mn} \cos m\phi$$

For $x \ll 1$, $I_m \rightarrow \frac{1}{\Gamma(m+1)} \left(\frac{x}{2}\right)^m$ (3.102)

$$\Phi(\rho, \phi) = \frac{16V}{\pi^2} \sum_{m \text{ odd}} \sum_{n \text{ odd}} \frac{\frac{1}{\Gamma(m+1)} \left(\frac{\rho}{2L}\right)^m}{\frac{1}{\Gamma(m+1)} \left(\frac{b}{2L}\right)^m} (-1)^{\frac{n-1}{2}} \frac{(-1)^{m-1/2}}{mn} \cos m\phi$$

$$= \frac{16V}{\pi^2} \sum_{m \text{ odd}} \frac{\rho^m}{b^m} \frac{(-1)^{\frac{m-1}{2}}}{m} \cos m\phi \underbrace{\sum_{n \text{ odd}} \frac{(-1)^{\frac{n-1}{2}}}{n}}_{= \frac{\pi}{4} \text{ alternating odd harmonic series}}$$

$$= \frac{4V}{\pi} \sum_{m \text{ odd}} \frac{\rho^m}{b^m} \frac{(-1)^{\frac{m-1}{2}}}{m} \cos m\phi$$

Investigating problem 2.13 & $\Phi(\rho, \phi) = \frac{2V}{\pi} \tan^{-1} \left(\frac{2b\rho \cos \phi}{b^2 - \rho^2} \right)$ for $V_1 = V$
 $V_2 = -V$

We see that these are very similar & indeed they are identical

Investigating the power series of $\tan^{-1}x$

$$\begin{aligned} \tan^{-1}x &= \int \frac{1}{1-x^2} dx \\ &= \int \sum_n (-1)^n x^{2n} dx \\ &= \sum_n (-1)^n \frac{x^{2n+1}}{2n+1} = \sum_{m \text{ odd}} (-1)^{\frac{m-1}{2}} \frac{x^m}{m} \quad \text{for } 2n+1 = m \end{aligned}$$

We can therefore use that to work with Φ

$$\Phi = \frac{4V}{\pi} \sum_{m \text{ odd}} \frac{\rho^m}{b^m} \frac{(-1)^{\frac{m-1}{2}}}{m} \cos m\phi$$

$$= \frac{4V}{\pi} \sum_{m \text{ odd}} \mathcal{R} \left[\left(\frac{\rho}{b} e^{i\phi} \right)^m \right] \frac{(-1)^{\frac{m-1}{2}}}{m}$$

$$\mathcal{R}(e^{im\phi}) = \cos m\phi \quad \text{De Moivre.}$$

$$= \frac{4V}{\pi} \mathcal{R} \left[\tan^{-1} \left[\frac{\rho}{b} e^{i\phi} \right] \right]$$

We use the identity $2\tan^{-1}z = \tan^{-1}\left[\frac{2z}{1-z^2}\right]$

$$\begin{aligned}\bar{\Phi} &= \frac{4V}{\pi} \operatorname{Re}\left[\tan^{-1}\left[\frac{2\rho_b e^{i\phi}}{1-\rho^2/b^2}\right]\right] \\ &= \frac{4V}{\pi} \tan^{-1}\left(\frac{2pb\cos\phi}{b^2-\rho^2}\right) \quad \text{as desired.}\end{aligned}$$